SOVEREIGN INDUSTRIAL FACILITY
INSPECTION REPORT #INSP-2026-0842

Location: Unit 4 - High-Pressure Boiler & Turbine Hall
Inspector: Senior Reliability Engineer J. Doe
Date: 2026-08-24
Status: REQUIRES ATTENTION

1. EXECUTIVE SUMMARY
Annual non-destructive testing (NDT) inspection conducted on Boiler Feed Pump BFP-02 and Main Steam Head

2. CRITICAL OBSERVATIONS & DEFECTS
- Observation A: Ultrasonic thickness gauging on elbow E-104 indicated wall thinning to 4.2mm (Design minimun
- Observation B: Vibration harmonics on Bearing #2 of BFP-02 exceed ISO 10816-3 limits (7.8 mm/s RMS). Poss
- Observation C: Pressure relief valve PRV-301 calibration is overdue by 45 days.

3. RISK CLASSIFICATION
- Elbow E-104: HIGH RISK (Risk of uncontained steam leak at 120 bar)
- Bearing BFP-02: MEDIUM RISK (Operational disruption risk)
- Valve PRV-301: HIGH RISK (Regulatory compliance non-conformance)

4. RECOMMENDED CORRECTIVE ACTIONS
1. Schedule immediate hot-tap replacement or de-rating of Elbow E-104 within 14 days.
2. Replace Bearing #2 on Pump BFP-02 during the upcoming planned outage.
3. Recalibrate and recertify PRV-301 immediately.